Ruiwen Wang Ph.D. candidate

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Nationality: French

Summary

Ph.D. (CIFRE) candidate in Computer Science at Sorbonne University & EURECOM, expected to graduate in Aug 2026, in collaboration with Huawei Paris Research Center. My research lies at the intersection of AI, HPC, and ML systems, with a focus on principled optimization-based planning for efficient and reliable ultra-scale LLM/DNN training. I develop symbolic performance and memory models together with ILP and constraint-based search to synthesize memory-feasible, communication-aware hybrid-parallel strategies and pipeline schedules. My work has been validated on production Ascend clusters up to 16,384 NPUs and informs deployable planning workflows, with publications at CLUSTER, Euro-Par (Best Poster Award), NPC, and HPC Asia.

Research Expertise

- Research on scalable LLM/DNN training systems: hybrid-parallel strategy design across tensor, data, pipeline, and expert dimensions, stage partitioning, and micro-batch scheduling.
- Principled modeling for system co-optimization: symbolic performance and peak-memory modeling, feasibility analysis, and validation methodology for large-scale clusters.
- Optimization-based planning algorithms: ILP and constraint programming for joint optimization of memory, communication, and throughput.
- Translation of systems research to practice: reproducible benchmarking and deployable workflows in collaboration with engineering and platform teams.

Professional Experience

Distributed and Parallel Computing Lab, Huawei Paris Research Center — Paris, France
Ph.D. Candidate (CIFRE) & Research Engineer – LLM Training Systems (expected) — 2021 – 2026

- Built cost-model-driven planners for hybrid/pipeline-parallel LLM training: stage partitioning, pipeline scheduling, and recomputation selection under feasibility (memory/performance) constraints.
- Designed symbolic peak-memory estimation and integrated collective-cost modeling with overlap strategies for large-scale throughput and MFU tuning.
- Translated prototypes into deployable planning workflows with engineering and customer teams: actionable configuration guidance, debuggable traces, and regression benchmarking across model families and scales.

Role progression: Research Apprentice (2021–2022) → Research Engineer (2022–2023) → CIFRE Ph.D. Candidate (2023–2026).

Publications

- **Ruiwen Wang**, Chong Li, Thibaut Tachon, Raja Appuswamy, Teng Su. *BMPipe: Bubble-Memory Co-optimization Strategy Planner for Very-Large DNN Training.* 27th IEEE International Conference on Cluster Computing (CLUSTER 2025).
- **Ruiwen Wang**, Chong Li, Raja Appuswamy, Yujie Yuan. *H2O: Holistic Hyperparameter Optimization for Large-Scale Deep Neural Network Training.* **Best Poster Award.** 31st International European Conference on Parallel and Distributed Computing (Euro-Par 2025).
- **Ruiwen Wang**, Chong Li, Hongxing Wang, Raja Appuswamy, Yujie Yuan. *ManuMatic: Strategy Injection for Robust Automatic Hybrid Parallelism in Distributed DNN Training.* 22nd IFIP International Conference on Network and Parallel Computing (NPC 2025).
- **Ruiwen Wang**, Chong Li, Thibaut Tachon, Raja Appuswamy. *SCOPE: Symbolic Computation-Memory Optimization for Pipeline Efficiency in Ultra-Scale DNN Training.* 1st International Workshop on Distributed and Parallel Programming for Extreme-scale AI (DP2E-AI 2025).
- Xinzhang Liu, Chao Wang, Zhihua Yang, Zhuo Jiang, …, **Ruiwen Wang**, et al. *Training Report of TeleChat3-MoE.* arXiv preprint, 2025.
- **Ruiwen Wang**, Philippe Fang, Chong Li, Thibaut Tachon, Raja Appuswamy. *PRISM: Profiling-Free Symbolic Memory-Driven Strategy Planner for Large DNN Model Training.* 19th Supercomputing Asia / International Conference on High Performance Computing in the Asia-Pacific Region (SCA/HPC Asia 2026).
- Zhengdao Yu, **Ruiwen Wang**, Nelson Lossing, Chong Li. *MLLM Pipeline Bubble Modeling for Large-Scale Training.* 19th Supercomputing Asia / International Conference on High Performance Computing in the Asia-Pacific Region (SCA/HPC Asia 2026).

Patents

- Chong Li, Pierre Leca, Thibaut Tachon, **Ruiwen Wang**, Haoran Wang. *Devices and methods for generating execution plans for a machine learning model.*
- Chong Li, **Ruiwen Wang**, Haoran Wang, Fan Yu. *A load balancing method to improve large model performance.*
- Chong Li, Thibaut Tachon, **Ruiwen Wang**, Haoran Wang. *A multi-dimension parallelism configuration method for large language model training.*

Education

Sorbonne University & EURECOM — Paris, France
Ph.D. in Computer Science — 2023 – 2026 *(expected)*
- Supervisor: Prof. Raja Appuswamy, Prof. Chong Li
- Research topic: Systematic and Portable Acceleration of Distributed Learning using Parallel Computing Techniques

Sorbonne University — Paris, France
M.Sc. in Computer Science — 2019 – 2022
- Supervisor: Prof. Emmanuel Chailloux & Prof. Jacques Malenfant
- Research focus: Algorithms, Software Science, and Technology

Paris-Saclay University — Paris, France
B.Sc. in Computer Science — 2016 – 2019

Awards and Honors

- **Golden Prize**, Huawei Challenge Award for Automatic Load Balancing — 2024.08
- **Best Poster Award**, Euro-Par conference — 2025.08
- **Letter of Appreciation**, China Telecom (DeepSeek-MoE training optimization deployment) — 2025.12

Skills

Languages: Chinese (native), English (fluent), French (fluent), Cantonese
Programming: Python, C++, Java
ML Frameworks: MindSpore, PyTorch
Platforms: large-scale GPU/NPU clusters and cloud
Tooling: Ascend profiling, tracing, performance regression benchmarking
Development: Linux, Git